

Screen-Shooting Robust Watermark Based on Style Transfer and Structural Re-Parameterization

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Abstract—In real-world applications, screen capturing represents a significant scenario where this process can induce substantial distortion to the original image. Previous methods for simulating screen-shooting distortion often involved combining different formulas. We found that these simulation methods still have a significant gap compared to real distortions, making it urgently necessary to develop a realistic and credible comprehensive noise layer to achieve robustness against screen-shooting distortion. This paper presents a watermarking scheme capable of withstanding severe screen-shooting distortion. First, a dataset is constructed to train a screen-shooting distortion simulation network based on style transfer. Subsequently, a comprehensive noise layer is built upon this network to achieve robustness against severe screen-shooting distortion. Additionally, this paper incorporates structural re-parameterization techniques into the traditional U-shaped encoder to improve the quality of encoded images. Extensive experiments demonstrate the proposed scheme's superior performance in terms of robustness and generalization, especially under severe screen-shooting distortion conditions.

Index Terms—Data hiding, robust watermark, screen-shooting simulation network, adversarial training, visual image codes.

I. INTRODUCTION

THE proliferation of electronic devices has made screen capturing a common scenario in daily life. When capturing a screen with a mobile phone, we often observe that the image may appear overly bright or dark, blurred, distorted, or exhibit irregular wavy interference patterns, all of which significantly degrade the image quality. Such screen-shooting

scenarios pose greater challenges to the physical robustness of images. Traditional digital watermarking techniques have traditionally focused on robustness against electronic channel distortions [1], [2], [3], such as JPEG compression, scaling, Gaussian noise, etc. Screen capturing severely disrupts the watermark signal of images, posing significant challenges for traditional digital watermarking techniques in extracting watermarks from screen-shooting images.

Recently, researchers have proposed a deep learning-based watermarking framework [4], [5], [6] comprising an encoder, decoder, and noise layer. The key to achieving distortion robustness lies in designing an effective noise layer trained to withstand specific types of noise. This approach is feasible in electronic channels where noise is typically differentiable. However, due to the non-differentiable nature of various physical distortions, such frameworks struggle to adapt to screen-shooting channels.

Fang et al. [6] proposed a novel approach wherein the entire process of quantitatively simulating screen-shooting noise layers is deemed unnecessary. Instead, simulating the most significantly affected distortions is sufficient to generate a robust noise layer. This method identifies perspective distortion, lighting distortion, and moiré distortion as the most influential distortions in the screen-shooting process, approximating them in a differentiable manner to enhance robustness against screen-shooting images. However, it was observed that the simulated distortions in this noise layer still exhibit significant disparities compared to real distortions. On the one hand, this manifests as relatively mild visual distortion. On the other hand, screen-shooting distortions arise from a complex and irregular blend of multiple distortion types mixed in specific proportions, rendering Fang's approach of using additive differentiable formulas for simulation unreasonable.

There are two commonly used methods for constructing noise layers: one is to approximate simulation using differentiable mathematical formulas, and the other is to train networks for simulation. For the former, screen-shooting processes involve various irregular distortions in appearance, which are challenging to simulate using mathematical methods. For the latter, screen-shooting scenarios have received less attention in the past, resulting in a scarcity of datasets related to screen-shooting images. Furthermore, although current watermarking schemes have achieved commendable performance, they still face challenges in balancing robustness and imperceptibility. Addressing these challenges by optimizing the underlying

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network architecture more effectively embed watermarks while maintaining or even improving the quality and performance of the watermarked content remains a pressing issue that requires resolution.

To address the issues above, a novel and robust watermarking scheme is proposed. This scheme is capable of extracting watermarks from screen-captured images even under substantial physical distortion interference. We first construct a dataset of screen-shooting distortions, comprising original images paired with corresponding real screen-shooting distorted images. Subsequently, a screen-shooting image simulation network based on style transfer is designed and trained using this dataset. Finally, an end-to-end robust watermarking scheme is proposed based on the screen-shooting image simulation network, consisting of an encoder, a decoder, and a comprehensive distortion layer. The encoder employs structural re-parameterization techniques to enhance the invisibility of watermark images. This approach optimizes the network's structure, allowing for more efficient representation learning and reducing artefacts that may arise during the encoding process. The comprehensive distortion layer integrates the screen-shooting image simulation network. It is jointly trained with the encoder and decoder, ultimately achieving strong robustness against screen-shooting distortions while maintaining good image quality. Through this enhancement, the study aims to push the boundaries of current image encoding methodologies and offer a more robust solution for high-quality image reconstruction tasks.

The proposed approach contributes as follows:

- 1) For the first time, this study employs a network to simulate screen-shooting distortion layers, effectively using style transfer-based networks to generate high-quality screen-shooting distortion images. To train this network, we constructed a large dataset comprising 10,000 images with diverse screen-shooting distortions from different distances and angles.
- 2) This study incorporates structural re-parameterization techniques into the traditional U-shaped encoder architecture to improve the visual quality of the encoded images. By integrating these advanced techniques, the model is able to capture better and preserve fine details and textures, resulting in more accurate and visually appealing outputs.
- 3) This paper proposes an end-to-end watermark extraction network based on a screen-shooting distortion simulation network consisting of an encoder, decoder, and noise layer, where the screen-shooting distortion simulation network is integrated as a crucial component of the noise layer. Extensive real-world experiments demonstrate that the proposed model excels in robustness and generalization, maintaining outstanding performance across various datasets and electronic devices.

II. RELATED WORK

The robust watermarking scheme is categorized into two main channels: electronic channels and screen-shooting channels. The electronic channels pertain to the direct embedding and extraction of watermarks within digital media, ensuring

high fidelity and minimal distortion. On the other hand, the screen-shooting channels involve capturing the watermarked content through a camera or screen-capture process, which can introduce various distortions and require specialized techniques for accurate watermark extraction. As the capabilities of cameras on smartphones, tablets, and other devices continue to improve, the transmission of screen content through photographs or screenshots has become increasingly common. This trend underscores the growing urgency of protecting digital content's copyrights and security.

A. Robust Watermarking Scheme Under Electronic Channels

Traditional robust image watermarking algorithms [7], [8], [9] primarily focus on distortions in electronic channels, embedding watermark information by modifying image pixels. Inspired by the success of CNNs in the field of computer vision, several deep learning-based digital image watermarking methods have emerged.

Ahmadi et al. [10] applied a predefined embedding algorithm in the feature maps of autoencoders, leveraging CNN's powerful fitting capability to transform image features into the transform domain. Intensity factors were introduced into the encoder to balance robustness and imperceptibility flexibly.

Liu et al. [11] introduced a two-stage deep learning robust watermarking model. Unlike previous watermarking models, this model separates the embedding and extraction of watermarks into two stages, offering the advantage of not requiring noise attacks to be modeled as differentiable noise layers. This enables the decoder to withstand better noise attack types that are challenging to model directly as differentiable network layers, such as JPEG compression.

Fei et al. [12] proposed a watermarking method to protect intellectual property in generative adversarial networks by embedding an invisible watermark into GAN models. This approach ensures that any image generated by the GAN contains an imperceptible watermark, allowing for ownership verification through watermark detection. Fei inserted a pre-trained CNN watermark decoding block at the output of the generator and modified the generator loss by including a watermark loss term to ensure the specified watermark could be extracted from generated images.

Whether based on the spatial domain or transform domain, digital watermarking algorithms can be tailored to resist specific image processing distortions as needed, enabling the design of watermark embedding schemes robust against common image processing operations.

B. Robust Watermarking Schemes Under Screen-Shooting Channels

The widespread of electronic devices has made screen-shooting scenarios a common occurrence in daily life. Robust watermarking schemes under screen-shooting channels have become a significant research challenge.

Tancik et al. [13] proposed an end-to-end deep learning framework called StegaStamp, which embeds arbitrary hyperlinks into natural colour images in a nearly perceptually

invisible manner, successfully decoded after printing and digital display.

Based on StegaStamp, Jia et al. [14] introduced a novel approach where a distortion network utilizes differentiable 3D rendering operations to simulate distortions occurring in print and display scenarios. Additionally, using loss functions aligned with the Human Visual System (HVS) to supervise encoder training maintains the visual appeal of the images.

Fang et al. [6] and colleagues argued that simulating only the most affected distortions is sufficient to generate an effective noise layer robust to screen-shooting, proposing a screen-shooting noise layer named PIMoG. They also introduced gradient mask-guided image loss and edge mask-guided image loss further to enhance the robustness and invisibility of the entire network.

Liu et al. [15] presented the WRAP watermarking scheme, pioneering the study of digital watermarks in film coating scenarios, demonstrating robustness on film-coated prints across various datasets, film types, and cameras. They also proposed a perceptual image hashing scheme based on the Swin Transformer to resist the impact of screen capture distortion on content authentication [16]. During training, a distortion simulator was added, which used image processing operations to simulate distortions on the original image. This aided the feature extractor and hash generator in learning to resist screen capture distortions.

Huang et al. [17] proposed ARWGAN, which incorporates a Feature Fusion Module (FFM) to improve robustness through dense connections. Additionally, an Attention Module (AM) is employed to emphasize inconspicuous and textured regions for watermark embedding, thereby optimizing the watermarking performance.

Xiao et al. [18] addressed the issue of computational and communication burdens affecting system scalability by designing a pair of image encryption and personalized decryption algorithms based on matrix operations. They achieved this by transferring the watermark embedding operation to the screen-decrypted image.

He et al. [19] introduced an image-instance-level robust watermarking scheme, referred to as CSRW-IIL, comprising a template embedding module at the encoding end and a geometric correction module at the decoding end. The former embeds templates with adaptive magnitude correlation perception multiplication spreading. At the same time, the latter learns a calibrator to estimate perspective projection, proving effective against various camera capture attacks under diverse shooting conditions, regardless of whether the entire display image is captured.

Overall, the research on robust watermarking techniques for screen-shooting scenarios is continually advancing, yet many challenges remain to be addressed. Future research will focus on developing more robust, efficient, and cross-device-compatible watermarking schemes to meet the increasingly complex demands of digital content protection.

III. PROPOSED WATERMARKING NETWORK MODEL

In this paper, we propose a novel robust watermarking model based on Generative Adversarial Networks (GANs). In

this section, we first provide an overview of the framework of the watermarking model, followed by a detailed description of each component.

A. Methodology Overview

As shown in Fig. 1, the proposed model mainly consists of an encoder E_α , a decoder D_β , a noise layer N , and a discriminator C . Here, α and β represent the hyperparameters corresponding to the encoder and decoder. These are used during the training process to optimize the training outcome, enhancing both invisibility and robustness. The watermark information W_{in} of length L is first processed into a message tensor T_m of the same dimension as the encoder's input. The original image $I_{in} \in \mathbb{R}^{H \times W \times C}$ and the message tensor T_m are input into the encoder to generate the encoded image I_e . The discriminator C calculates the similarity between the original image I_{in} and the encoded image I_e with the aim of improving the quality of the encoded image I_e generated by the encoder. The decoder is used to extract the watermark information from the noised image I_n and to ensure that the decoded watermark information W_{de} is as consistent as possible with the input watermark information W_{in} .

B. Message Pre-Processing Layer

Since the dimension of the binary watermark information sequence $W_{in} \in \{0, 1\}$ with length $L = 30$ does not conform to the encoder's input dimension, it needs to be processed by the information preprocessing layer H_m first. This layer is mainly composed of a fully connected layer, which takes the received watermark information W_{in} as input and then generates a vector of length 256. The vector is reshaped and replicated to generate the message tensor $T_m \in \mathbb{R}^{64 \times 16 \times 16}$, which is subsequently up-sampled to align with the dimensions of the input layers of the encoder, as illustrated in Fig. 1.

C. Pretrained Edge Mask Detector

Incorporating a significant weight to the image's edge region during watermark signal embedding has been shown to enhance the overall visual appearance of the resulting watermarked image [20]. We introduced a mask-guided image loss to achieve this automatically within a deep-learning framework. The acquisition of the mask is facilitated by the utilization of a previously trained edge detection network known as BDCN [21], which has effectively captured image edges. With the edge mask $M_{edge} \in \{0, 1\}$ generated, the loss function for the encoded image can be defined as follows:

$$\mathcal{L}_I = M'_{edge} \times MSE(I_{in}, I_e) \quad (1)$$

where

$$M'_{edge} = M_{edge} \times \alpha_E + \beta_E \quad (2)$$

variables α_E and β_E are utilized to modify the weights of the mask. This paper establishes the default value of α_E and β_E are set as 1. M_{edge} ensures that the lower value is kept in the edge region, whose main function is to guide the encoder to increase the distribution weight of the edge area when embedding the watermark.

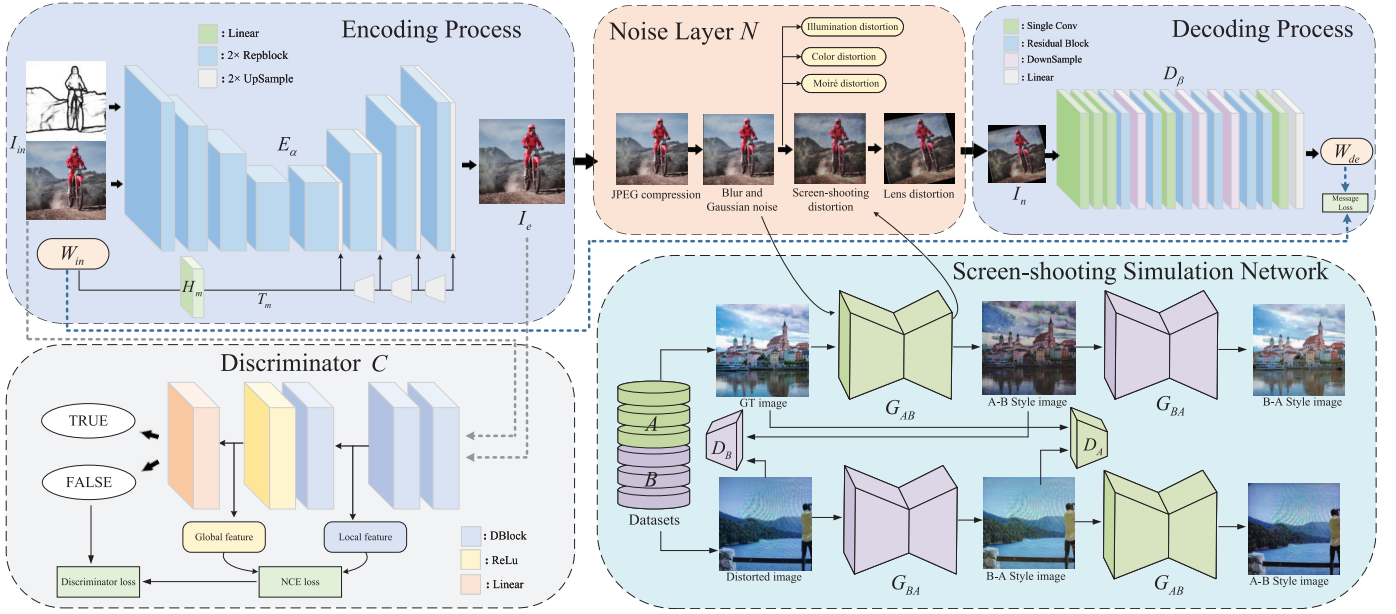


Fig. 1. Overall architecture. In the proposed model, the original image is inputted into the encoder to obtain the encoded image. The encoded image is displayed on the screen, allowing the user to capture it again using a camera. Once the recaptured encoded image is obtained, it is fed into the decoder, extracting the embedded watermark information from the image.

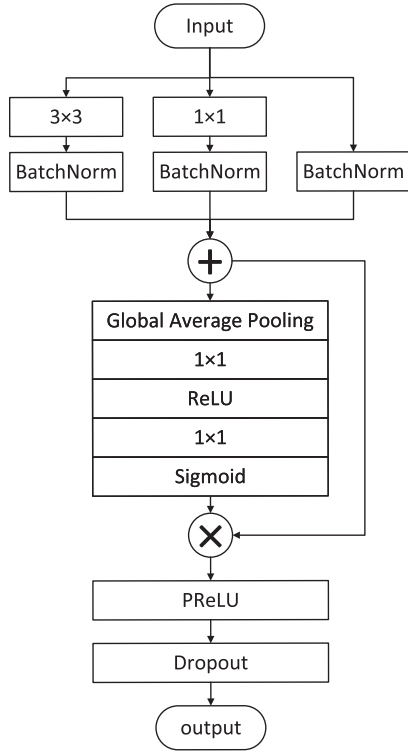


Fig. 2. Repblock architecture.

D. Encoder

The structure of the encoder is shown in Fig. 1. Its main task is to combine the message tensor T_m with the original image I_{in} to generate a watermarked image I_e that is imperceptible to human eyes. The process can be expressed by the following formula:

$$I_e = E(\alpha, I_{in}, T_m) \quad (3)$$

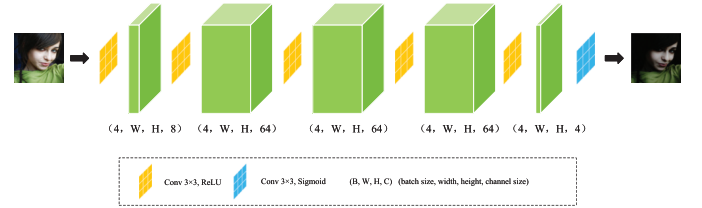


Fig. 3. Illumination reduction module structure.

where $E(\cdot)$ represents the encoder, α is the hyperparameter of the encoder.

In this paper, structural re-parameterization techniques [22] are introduced into the traditional U-shaped encoder structure to expand the receptive field and improve the quality of the image after watermark embedding. We refer to the reparameterization module as the “Repblock,” and its structure is shown in Fig. 2. It mainly consists of a 3×3 convolution, a 1×1 branch, and a shortcut branch. The results of the three branches are batch-normalized and summed. This result is then multiplied by the result of the global average pooling- 1×1 convolution-ReLU- 1×1 convolution-Sigmoid block, and the final output is obtained after PReLU and Dropout operations.

The overall encoding process can be represented explicitly as follows: the input image is first downsampled to a $(H \div 8) \times (W \div 8)$ feature map using three Repblocks. This feature map is then concatenated with the global feature map, obtained by global average pooling and a 1×1 convolution. After concatenating the two feature maps, three upsampling layers are applied to embed the watermark information into the feature map, ultimately generating a watermarked image with the exact dimensions as the original image.



Fig. 4. Comparison of distortion simulation results among different schemes and distortion in real scenes.

E. Screen-Shooting Simulation Network

In our experiments, we found that there is a specific connection between moiré patterns and illumination noise in reality, which cannot be achieved by simply simulating them separately and then adding them together. This is because the moiré effect is essentially the mechanical interference of light by superimposed networks of lines [23]. Also, color distortion caused by shooting, such as chromatic aberration, is also caused by dispersion: the refractive index of the lens elements varies with the wavelength of light [24]. Considering the mutual influence of noise in actual screen shooting scenes based on physical principles, it is inaccurate to simulate it by simply superimposing several different types of noise. Most previous solutions used mathematical formulas to simulate illumination noise, moiré noise, and color distortion separately without considering the intrinsic connection between them. Therefore, we use a Generative Adversarial Network (GAN) to learn the style characteristics of authentic captured images, aiming to simulate noise that more closely matches reality. We took 10,000 screen-shooting distorted images in real scenes as a training set and sent them into the style transfer network CycleGAN [25] for training.

As shown in Figure 1, in the screen-shooting distortion network, the data is divided into two domains: Domain A and Domain B, where Domain A represents the ground truth images and Domain B represents the actual screen-shooting distorted images. The generator G_{AB} is responsible for converting the style features of images in Domain A into the style features of those in Domain B, and then the discriminator D_B evaluates the resulting images. Conversely, the generator G_{BA} transforms the style features of images in Domain B back into the style features of those in Domain A, and then the discriminator D_A evaluates the resulting images. Regarding the specific structure of the discriminator, refer to Subsection III-H below. Through adversarial training between the generators and discriminators, the quality and authenticity of the generated images are refined, resulting in

outputs that closely approximate the target domain's style. After that, we obtained a network that could simulate actual screen noise very well. The simulation effect is shown in Fig. 4. The simulation results are quite impressive across various subjects, including animals, people, and landscapes. Through the diversity of the style transfer network, we are able to simulate screen-shooting noise while keeping the connection in physics among moiré patterns, illumination noise, and color distortion, making it more consistent with the real environment. Afterward, the trained screen-shooting simulation network is added to the noise layer for decoder robustness training.

In addition, we have designed an illumination reduction module within the structure of the generator G_{AB} , which is positioned prior to the output of the G_{AB} . This module simulates the brightness reduction problem caused by screen shooting in real-world scenarios. Its structure is depicted in Fig. 3. For the actual simulation results, please refer to the Appendix.

F. Noise Layer

The robustness of the watermark model mainly relies on the noise layer. The decoder obtains the ability to resist specific distortions by correctly decoding the watermark information contained in the distorted image. In addition to the above-mentioned screen-shooting simulation network, the noise layer also includes a series of other distortions less related to light effects, which will be described in detail below.

1) *Lens Distortion*: During photography, the diverse positions from which humans take pictures can lead to multi-angle shots, resulting in RST (Rotation Scaling Translation) distortion after lens transformation. Lens distortion encompasses various geometric distortions, including rotation, cropping, deformation, and lens transformation. Lens distortion is particularly evident after lens correction and can lead to losing image details, which is present in every screen-shooting image. The proposed model simulates lens transformations using the

following formula:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = A \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad (4)$$

where A is a random homography matrix, $(x, y, 1)$ represents the coordinates of the untransformed encoded image. (X, Y, Z) means lens-transformed coordinates of encoded image. In this paper, the four corners of the encoded image are perturbed randomly (up to ± 40 pixels), and new coordinates (X, Y, Z) for each corner are generated, which together with Eq. (4) are used to find A utilized to calculate other lens transformed coordinates.

2) *Motion and Defocus Blur*: For motion blur, we use a convolution kernel with a fixed size between 3 and 7 to slide on the original image. At each position, we calculate the mean of the pixels covered by the convolution kernel and replace the pixel value of the element corresponding to the center point of the convolution kernel with the calculated mean. Similarly, for defocus blur, we do not calculate the mean of the pixels but instead calculate the Gaussian distribution, and the convolution kernel size is set between 1 and 3.

3) *JPEG Compression*: JPEG compression is a common image processing method used daily. It can significantly reduce the size of image files, saves storage space and transmission bandwidth, and the compressed image still maintains good visual quality. For the simulation of JPEG compression, we use Shin's scheme [26], which mainly solves the non-differentiable problem in the JPEG compression process. Shin closely approximates it using only differentiable operations:

$$\lfloor x \rfloor_{\text{approx}} = \lfloor x \rfloor + (x - \lfloor x \rfloor)^3 \quad (5)$$

where $\lfloor \cdot \rfloor$ represents rounding to the nearest integer, and x is the input image for JPEG compression.

4) *Gaussian Noise*: For other distortions that were not simulated, Gaussian noise was used to approximate their effects with parameter σ set to 0.02.

5) *Screen-Shooting*: The most important part of the noise layer is the screen-shooting simulation network. In the experiment, we found that if the simulated screen-shooting noise is too intense, it may lead to poor training results, so we set a hyperparameter p to adjust the simulation intensity of the screen-shooting network. In this paper, the value of p is 1. In summary, the noise layer can be summarized as follows:

$$I_n = N_l(p * N_s(N_b(N_j(I_e)) + N_g)) \quad (6)$$

where N_l represents the lens distortion, N_s denotes the screen-shooting simulation network, N_b represents the motion and defocus blur, N_j denotes the JPEG compression noise, N_g represents Gaussian noise.

G. Decoder

Decoder D is illustrated in Fig. 1. The major objective of this process is to extract the message from I_n . D comprises five single-convolution (conv-bn-relu-maxpool) blocks, nine residual blocks, and one linear block. Within the residual blocks, three downsampling operations are executed. The decoder loss

$\mathcal{L}_D$ is aimed to minimize the discrepancy between the decoded message, denoted as W_{de} , and the original watermark, denoted as W_{in} , by changing the parameter β :

$$\mathcal{L}_D = MSE(W_{in}, W_{de}) = MSE(W_{in}, D(\beta, I_n)) \quad (7)$$

H. Discriminator

In this paper, SNGAN [27] implements the discriminator. For the discriminator C and the encoder E , loss functions $\mathcal{L}_C$ and $\mathcal{L}_E$ can be denoted as:

$$\mathcal{L}_C = \gamma_1 \mathcal{L}_{nce}(X_r) + \mathcal{L}_{gan}(C, \hat{E}) \quad (8)$$

$$\mathcal{L}_E = \gamma_2 \mathcal{L}_{nce}(X_w) + \mathcal{L}_{gan}(\hat{C}, E) \quad (9)$$

where γ_1 and γ_2 are hyperparameters, both set 0.2 as default; X_r and X_w represent image sets from raw and watermarked images, respectively; $\hat{C}$ and $\hat{E}$ represent a fixed discriminator and encoder. Specifically, the discriminator and encoder are trained in an alternating manner. When calculating the discriminator loss $\mathcal{L}_C$, the raw image is provided as input, but the encoder remains unaltered. When calculating the encoder loss $\mathcal{L}_E$, the watermarked image is provided as input, while the discriminator remains unaltered. $\mathcal{L}_{nce}$ represents NCE loss [28]. $\mathcal{L}_{gan}$ in the above formula uses hinge loss [29]:

$$\begin{aligned} \mathcal{L}_{gan}(C, \hat{E}) &= \mathbb{E}_{x \sim p_r} [\min(0, 1 - C(x))] \\ &\quad + \mathbb{E}_{z \sim p_z} [\min(0, 1 + C(\hat{E}(z)))] \end{aligned} \quad (10)$$

$$\mathcal{L}_{gan}(\hat{C}, E) = -\mathbb{E}_{z \sim p_z} [\hat{C}(E(z))] \quad (11)$$

where p_r is the real data distribution, p_z is the prior noise distribution, and $E(z)$ is the watermarked image generated from sampling some random noise z .

I. Training Loss

The overall loss function comprises the previously described encoder loss $\mathcal{L}_E$, discriminator loss $\mathcal{L}_C$ and decoder loss $\mathcal{L}_D$, which can be stated as:

$$\mathcal{L} = \lambda_1 \mathcal{L}_E + \lambda_2 \mathcal{L}_C + \lambda_3 \mathcal{L}_D \quad (12)$$

where $\lambda_1, \lambda_2, \lambda_3$ are set as 0.5, 0.001, 3 by default.

IV. EXPERIMENTAL RESULTS

This section provides a comprehensive description of the experimental setup for our proposed method and presents the corresponding experimental data.

A. Dataset and Experimental Setting

Our training dataset, encompassing 15,000 images, has been randomly selected from the extensive COCO Train Images dataset [30]. Upon entering the testing stage, a well-chosen subset comprising 500 images was extracted randomly from the COCO Test Images dataset to facilitate the conduction of experimental evaluations. The underlying architecture of our system has been meticulously designed and implemented using the powerful PyTorch [31] framework, which subsequently operates on a GPU configured with the specifications of an NVIDIA RTX 3070. The experiments have been executed on

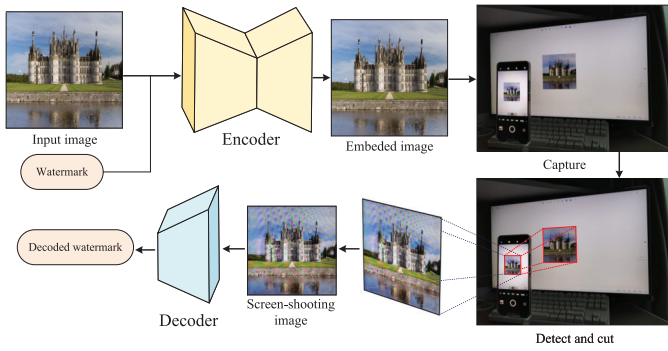


Fig. 5. Experimental flow chart.

TABLE I
THE PSNR AND SSIM VALUES OF EACH METHOD

Method	PSNR(dB)	SSIM
StegaStamp [13]	28.714	0.890
RIHOOP [14]	31.377	0.931
PIMoG [6]	37.141	0.972
WRAP [15]	29.554	0.880
ARWGAN [17]	30.059	0.920
CSRW-IIL [19]	33.118	0.948
Proposed	38.205	0.984

a display unit: the AOC LV273HUPR, featuring respective resolutions of 4k, providing optimal visualization conditions. The imaging equipment utilized during the capture process consists of the iQOO Neo9S Pro.

B. Implementation Details

The dimensions of training images are adjusted to $128 \times 128 \times 3$ pixels. The length l of the random message M is set to 30. The Adam optimizer [32] with default hyperparameters is employed for parameter optimization. The network model is subjected to 100 training epochs, with each epoch consisting of a batch size set to 32. During the robustness evaluation phase, we employed the abovementioned watermarked image region detection technique. The specific process of the experiment is shown in Fig. 5.

C. Visual Quality

In this subsection, we compare the visual quality of the proposed scheme with several classic and state-of-the-art schemes, and the outcomes are presented in Table I and Fig. 6. The baselines we have selected are from top-tier conferences and journals in the relevant fields. The quality of their works is very high, making them more representative of benchmark experiments. To evaluate the visual quality of the image that has been subjected to watermarking, we employ PSNR and SSIM as the standard assessment metrics. We randomly selected 5000 images from the COCO test set for image quality testing. Table I clearly demonstrates that the suggested model achieves the highest PSNR and SSIM among these schemes,

TABLE II
EXTRACTION ACCURACY UNDER DIFFERENT DISTANCES

Model	Distance	20cm	25cm	30cm	35cm	40cm
StegaStamp [13]		95.021%	97.229%	97.063%	97.167%	97.146%
RIHOOP [14]		95.833%	96.996%	96.917%	96.875%	96.813%
PIMoG [6]		84.074%	97.139%	97.278%	95.861%	94.611%
WRAP [15]		76.558%	76.867%	75.833%	74.300%	75.333%
ARWGAN [17]		94.694%	95.472%	94.750%	91.778%	90.111%
CSRW-IIL [19]		95.333%	97.101%	94.944%	95.435%	91.780%
Proposed		98.389%	98.889%	98.056%	96.972%	98.222%

indicating that our scheme has achieved satisfactory visual quality. This is because we use structure reparameterization technology to expand the receptive field, which can better capture the details of the image and retain more image features, thereby improving the encoder's performance and the quality of the encoded image compared with other methods.

Fig. 6 provides a comparison of the encoded images generated by the proposed model and several different methods. Since the purpose of this research field is to make the encoded image containing the watermark as similar as possible to the original image, it may be difficult to distinguish the difference in details with the naked eye. Therefore, we enlarge the local details of the encoded image to facilitate the observation of the detailed differences in texture among the encoded images generated by different methods. The experimental results show that the texture of the proposed scheme is basically consistent with the original image, and compared with other methods, it is not easy to observe the texture difference, which indicates that the proposed method has better invisibility.

D. Analysis on Screen-Shooting Robustness

Considering the diverse types of distortions that can potentially degrade the precision of information retrieval in real-world scenarios, we opt for the average bit correct extraction rate as a benchmark for performance assessment during the robustness evaluation stage. Our comparative analysis compares our strategy against several state-of-the-art watermarking techniques. Other approaches adopt the parameter values corresponding to their highest extraction rates, as reported in their respective papers.

In conducting comparative experiments, we have recognized the absence of standardized metrics for assessing the strength of moiré patterns in existing work, which has led to deviations in experimental data from other schemes when compared to their reported values in the papers. To address this, we have ensured consistency by fixing the distance and angle of equipment, thereby guaranteeing that the intensity of moiré interference remains uniform across each set of comparative trials, thereby minimizing the influence of differing experimental environments.

1) *Robustness Against Distance*: This subsection is dedicated to investigating the influence of distance and its effect on the decoding precision of our model. To carry out this meticulous examination, we maintain the camera's position at

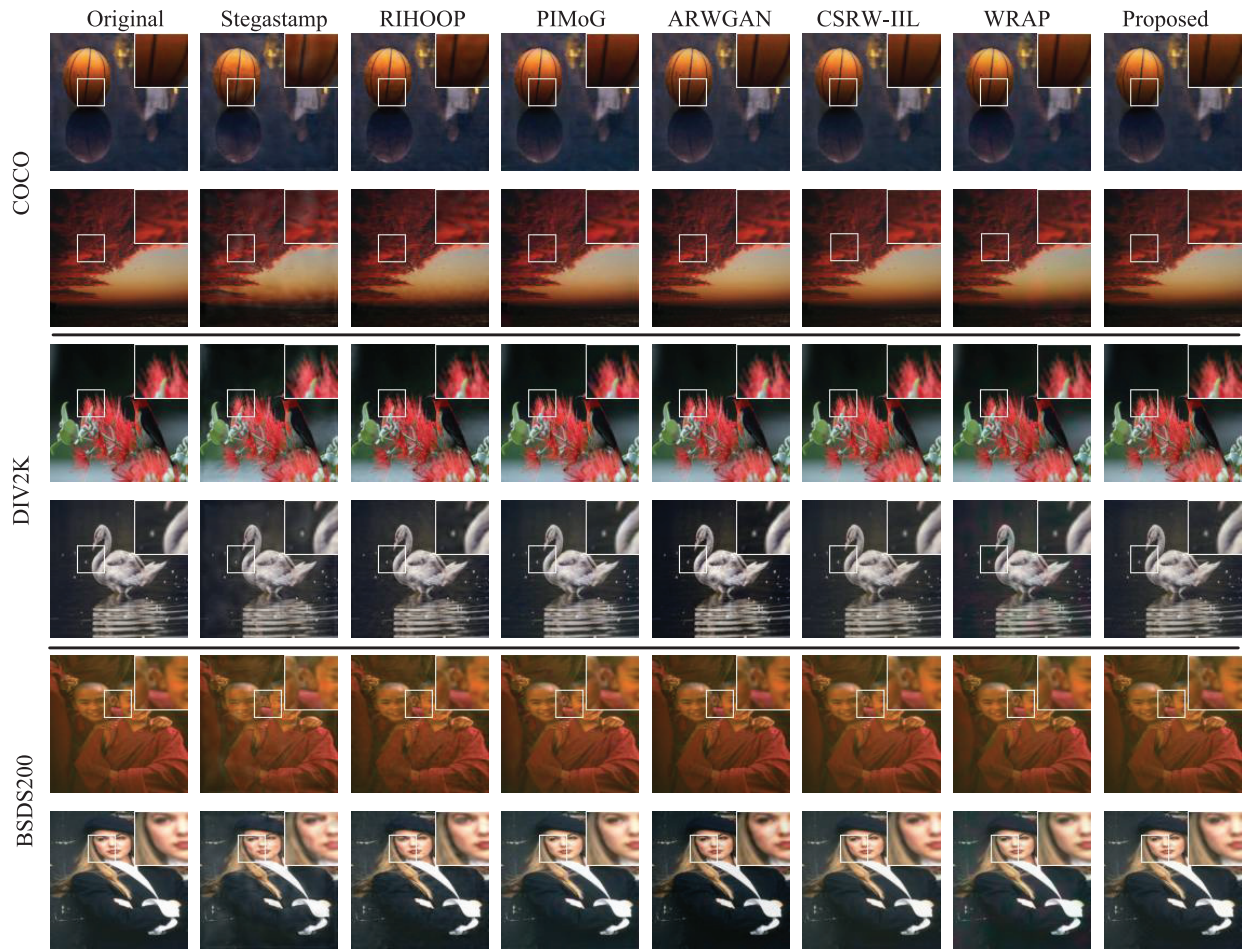


Fig. 6. Image visual quality among different methods.

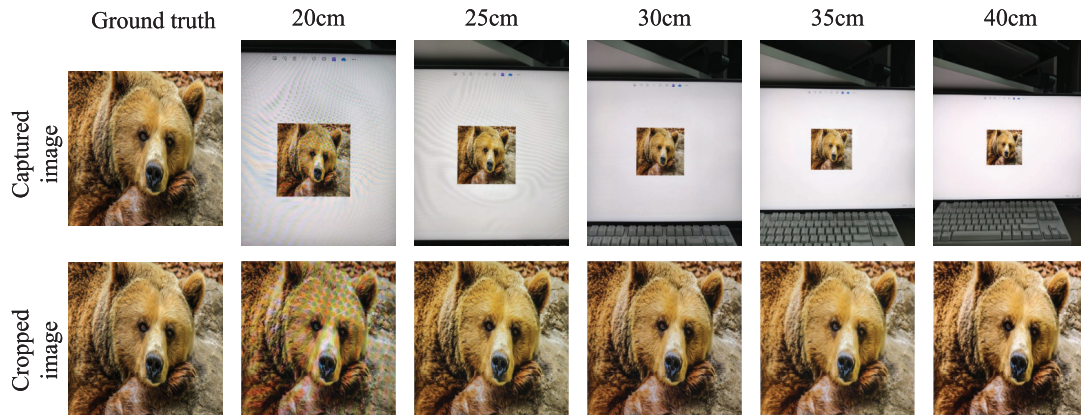


Fig. 7. Shooting environment and pictures at different distances.

a strictly perpendicular angle relative to the screen throughout the entire experimental process. The experimental results in Table II and Fig. 7 show that our model has good robustness at different distances. As can be seen from Table II, at all distances, the extraction rate of our model is above 96%, which is superior to other models. Fig. 7 intuitively shows the experimental environment when shooting at different distances, as well as the corresponding images used for decoding after removing the redundant edge areas after shooting. It can

be observed that when the distance is close, the moiré effect on the screen is more serious, and as the distance gradually increases, the moiré effect gradually decreases. This is because when the spatial frequency of the pixels of the shooting device is close to that of the screen pixels, interference phenomenon occurs between the pixels or sensors of the camera lens and the pixels or lines of the screen, and moiré patterns appear in the image. When the frequencies are offset, the severity of the moiré patterns will decrease. In addition, as the distance

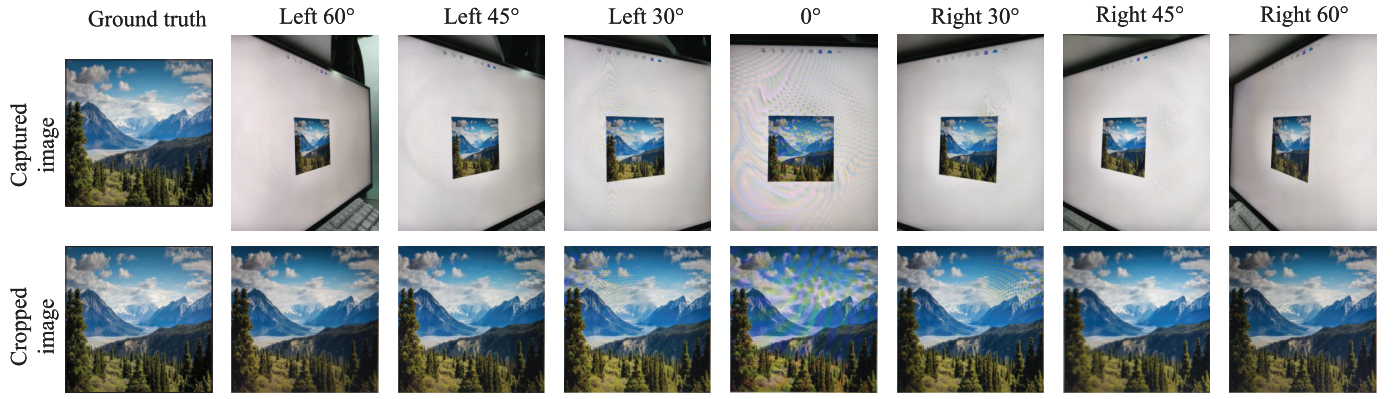


Fig. 8. Shooting environment and pictures at different angles.

TABLE III
EXTRACTION ACCURACY FROM VARYING ANGLES

Model \ Angles(°)	Left 60°	Left 45°	Left 30°	0°	Right 30°	Right 45°	Right 60°
StegaStamp [13]	96.438%	96.729%	97.000%	95.021%	97.146%	96.833%	96.396%
RIHOOP [14]	96.542%	97.042%	96.833%	95.833%	97.021%	96.813%	96.417%
PIMoG [6]	95.306%	95.861%	93.556%	84.074%	97.333%	93.861%	90.778%
WRAP [15]	73.467%	75.133%	76.767%	76.558%	76.583%	75.167%	71.125%
ARWGAN [17]	87.083%	93.667%	95.833%	94.694%	91.250%	89.417%	89.389%
CSRW-IIL [19]	95.238%	95.926%	94.278%	95.333%	94.218%	92.694%	94.389%
Proposed	96.694%	97.722%	97.056%	98.389%	98.250%	96.444%	96.835%

TABLE IV
EXTRACTION ACCURACY USING DIFFERENT METHODS WITH DIFFERENT DEVICES

Device	Method	20cm\0°	30cm\0°	20cm\30°	20cm\60°
Realme	StegaStamp [13]	96.938%	97.208%	96.896%	96.563%
	RIHOOP [14]	96.729%	97.042%	96.854%	96.604%
	PIMoG [6]	90.750%	96.139%	96.417%	91.222%
	WRAP [15]	76.375%	77.367%	76.342%	75.767%
	ARWGAN [17]	93.750%	95.889%	95.972%	90.250%
	CSRW-IIL [19]	97.139%	96.028%	93.556%	95.736%
	Proposed	99.667%	99.056%	98.222%	96.389%
HUAWEI	StegaStamp [13]	96.875%	97.104%	96.646%	96.000%
	RIHOOP [14]	96.833%	96.771%	96.688%	96.104%
	PIMoG [6]	93.250%	92.222%	90.833%	84.778%
	WRAP [15]	76.708%	77.775%	78.033%	76.333%
	ARWGAN [17]	73.472%	96.444%	95.966%	87.861%
	CSRW-IIL [19]	95.082%	96.610%	93.497%	95.464%
	Proposed	97.389%	98.306%	96.695%	96.214%

gradually increases, the resolution of the encoded image may be affected to a certain extent. Through the analysis of Fig. 7 and the data in Table II, in the case of the most severe interference, the proposed method achieves a better extraction rate and higher robustness.

2) *Robustness Against Spatial Distortion*: Our objective in this subsection is to examine how the decoding accuracy of our model is impacted by altering the shooting angle between the camera and the image. Camera pixel resampling due to

variations in shooting angles can interfere with information extraction. To ensure fairness in our experiments, our experimental setup was conducted at a fixed distance of 20 cm, given that we observed the most pronounced moiré interference at this 20cm distance.

Fig. 8 shows screen images taken at different angles and images used for decoding after angle correction. It can be seen from the figure that when the shooting device is facing the display device, the interference is the greatest. Table III

TABLE V
EXTRACTION ACCURACY USING DIFFERENT METHODS WITH DIFFERENT DATASETS

Datasets	Methods	20cm\0°	30cm\0°	20cm\30°	20cm\60°
BSD200	StegaStamp [13]	95.482%	97.438%	97.417%	97.250%
	RIHOOP [14]	96.186%	97.083%	97.354%	96.854%
	PIMoG [6]	77.423%	98.944%	97.361%	93.667%
	WRAP [15]	79.633%	77.850%	78.533%	76.333%
	ARWGAN [17]	84.889%	95.028%	95.361%	86.222%
	CSRW-IIL [19]	94.417%	98.361%	96.889%	92.333%
	Proposed	97.977%	99.806%	99.111%	96.444%
DIV2K	StegaStamp [13]	96.134%	97.458%	97.417%	97.250%
	RIHOOP [14]	95.913%	96.625%	96.292%	96.729%
	PIMoG [6]	80.526%	97.333%	96.694%	90.944%
	WRAP [15]	69.633%	74.542%	75.033%	73.333%
	ARWGAN [17]	89.278%	97.000%	97.056%	88.806%
	CSRW-IIL [19]	96.389%	96.639%	97.470%	92.056%
	Proposed	98.596%	100.000%	99.694%	98.528%

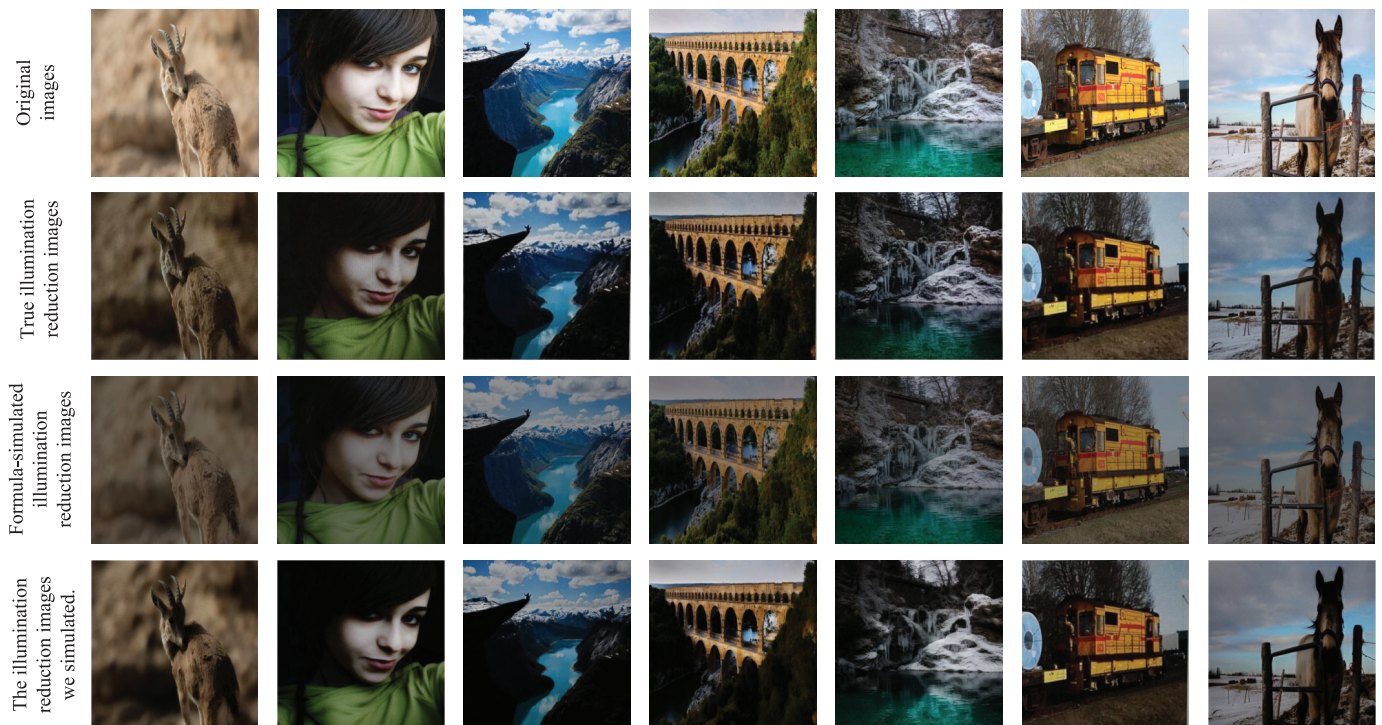


Fig. 9. Effect diagram of the illumination reduction module.

TABLE VI
MODEL PERFORMANCE WITH DIFFERENT MODULES

Repblock	Screen-Shooting Simulation Network	Accuracy	PSNR	SSIM
		49.056%	37.154	0.975
✓		50.741%	38.580	0.982
	✓	93.972%	37.738	0.979
✓	✓	98.389%	38.205	0.984

correspondingly presents the bit correct extraction rates for those respective angles. Experimental results show that, in most cases, the proposed scheme achieves the best extraction accuracy among these methods. Given the lower image quality

of the StegaStamp, even if it occasionally achieves a better extraction rate, it is generally inferior to the proposed solution. More pictures of the experimental environment are shown in the Appendix.

According to the above robustness experiments on distance and angle, the proposed screen capture simulation network has learned the interaction in physical principles among illumination noise, the moiré effect, and color distortion very well. It has achieved better results than the traditional method of separately simulating the two types of noise. In addition, the distortion pattern in the real scene is irregular, and the distorted image simulated by the screen capture simulation network is more consistent with the characteristic distribution of the original image and more realistic than the simulation using the formula.

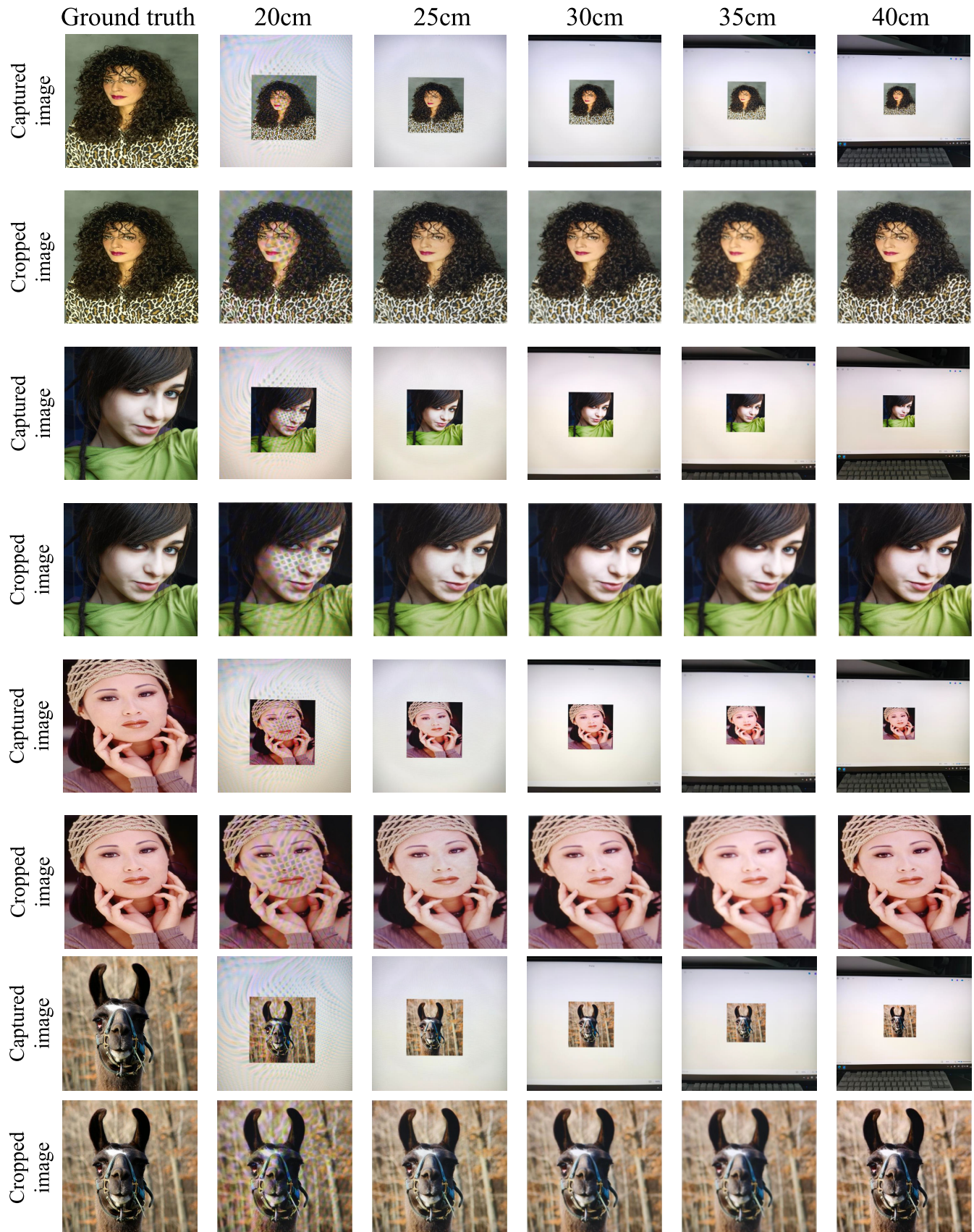


Fig. 10. Shooting environment and pictures at different distances.

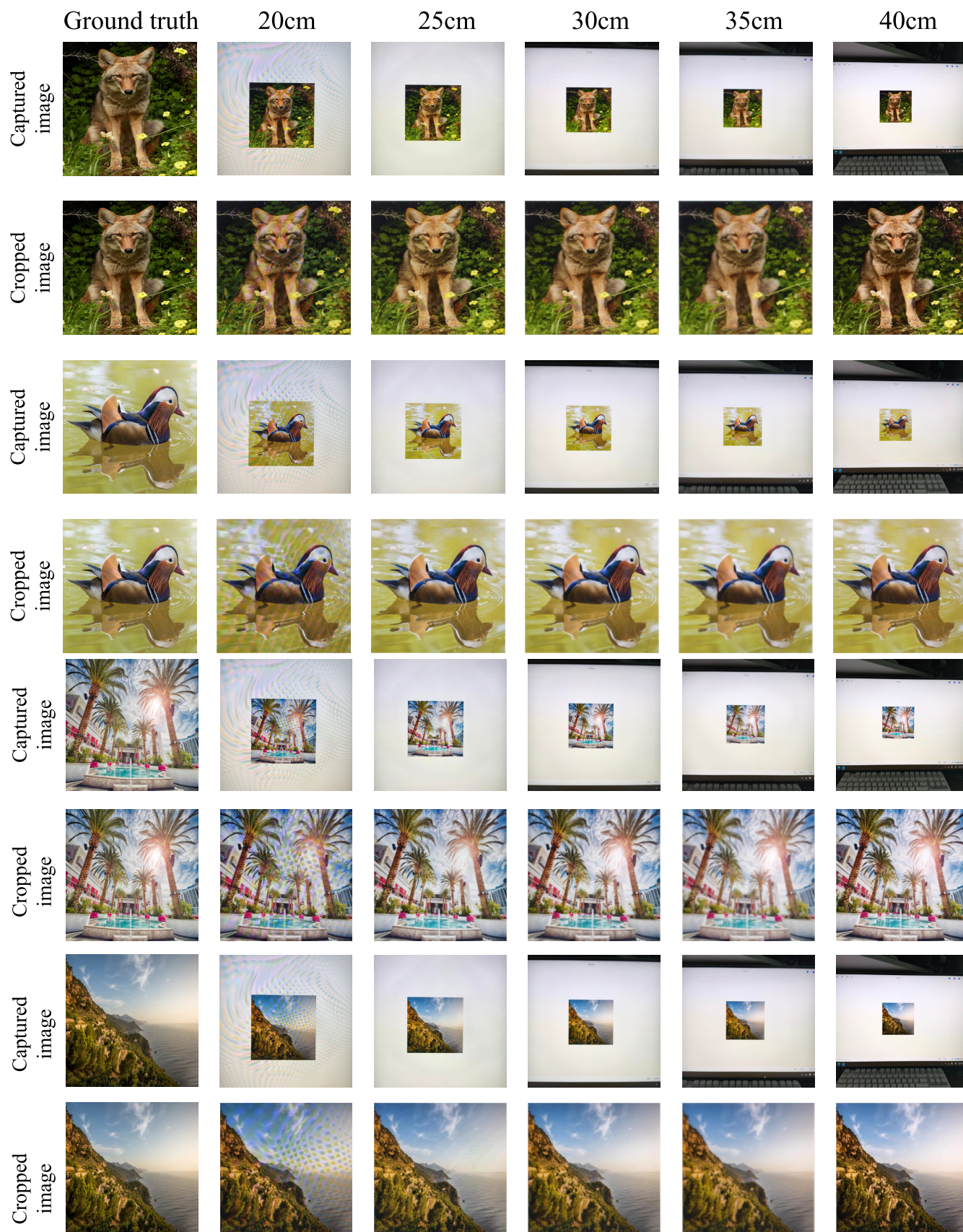


Fig. 10. (Continued.) Shooting environment and pictures at different distances.

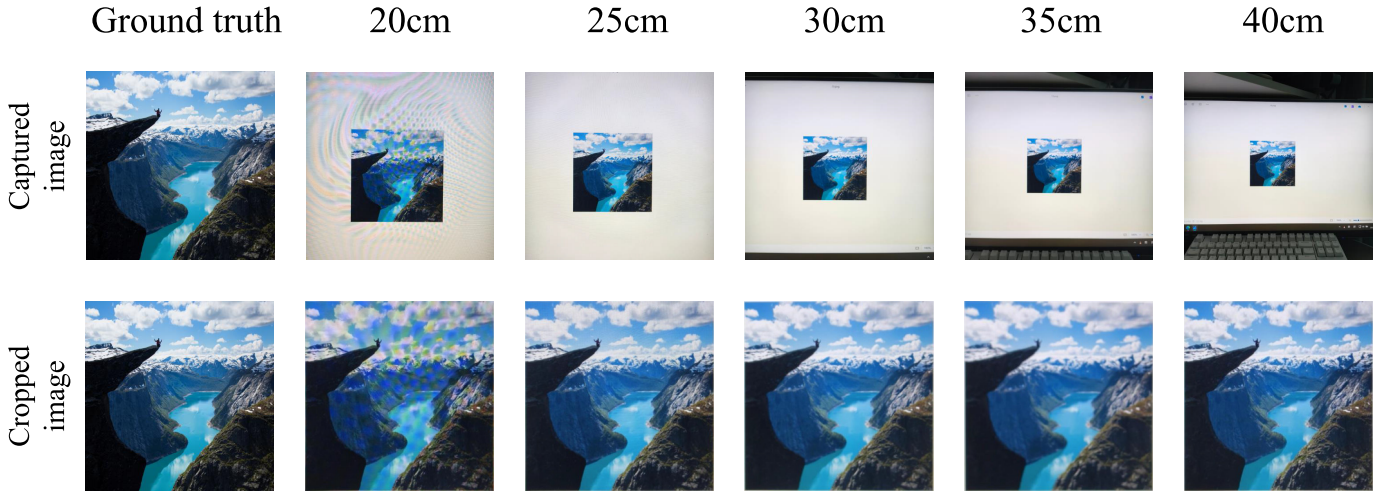


Fig. 10. (Continued.) Shooting environment and pictures at different distances.

E. Device Test

In practice, it is crucial to evaluate the applicability of the proposed method on various other devices. In order to evaluate the flexibility of the proposed method, we conducted a series of screen-shooting experiments using various types of capture devices. We use two different mobile phones (Realme RMX3366 and HUAWEI DBY-W09) for experiments, which is different from the phone used in other experiments. Since the moiré effect is determined by the frequency between the display device and the shooting device, there are slight differences in the moiré effect among different devices, resulting in slight fluctuations among experimental results. According to the data shown in Table IV, the proposed method is obviously superior to other methods for all device pairs in most cases, indicating the device adaptability of our proposed method.

F. Dataset Experiment

In order to prove the adaptability of the proposed scheme on the test set, in addition to the COCO test set, we selected two different test sets: BSD200 [33] and DIV2K [34], for performance testing, as shown in Table V. From the experimental data, we can see that the proposed scheme has achieved better performance in terms of extraction rate. For the StegaStamp scheme, although a set of data has achieved a good extraction rate, its PSNR and SSIM are not good, so the proposed scheme is still the best among all the compared schemes. Experimental data show that the proposed scheme is generalizable in different test sets, proving the effectiveness of the proposed scheme.

G. Ablation Experiment

In this part, ablation studies were conducted to verify the effectiveness of our proposed encoder with Repblock and screen-shooting simulation network in improving performance. We conducted a total of 4 sets of tests, including no module added, single module, and two modules combined test, to observe their image quality and extraction rate.

Table VI shows the extraction rate when the shooting angle is 0° and the shooting distance is 20 cm. Experimental data show that Repblock can improve the quality of encoded images, while the screen-shooting simulation network can improve the extraction rate of the model, that is, robustness. Ablation experiments fully demonstrate the effectiveness of the proposed module and explain the superiority of the proposed scheme in terms of robustness and image quality.

V. CONCLUSION

This paper proposes an end-to-end image watermarking method that is highly robust to screen-shooting noise. We constructed a dataset of 10,000 photos taken by mobile phones in real scenes. We used this dataset for style transfer training, aiming to allow the screen-shooting simulation network to learn the characteristics of screen-shooting noise in real scenes. In addition, we also optimized the architecture of the traditional U-style encoder and combined it with the structural reparameterization technology to improve the encoding ability of the model, thereby enhancing the quality of the encoded image. The proposed architecture is mainly jointly trained by the encoder, decoder, and discriminator, and the noise layer is used to improve the robustness of the model to specific noise. The trained model can correctly extract watermarks in the face of screen-shooting noise while ensuring image quality. A large number of experiments in real scenes show that the proposed method is superior to the state-of-the-art method in the screen-shooting scene test and has strong generalization ability in the face of different devices and test sets, which proves the superiority of this method.

APPENDIX

RELEVANT SUPPLEMENTARY MATERIALS

Due to space limitations, this section provides additional images from the experiments of the illumination reduction module, the distance and angle, as shown in Fig. 9, Fig. 10 and Fig. 11.



Fig. 11. Shooting environment and pictures at different angles.

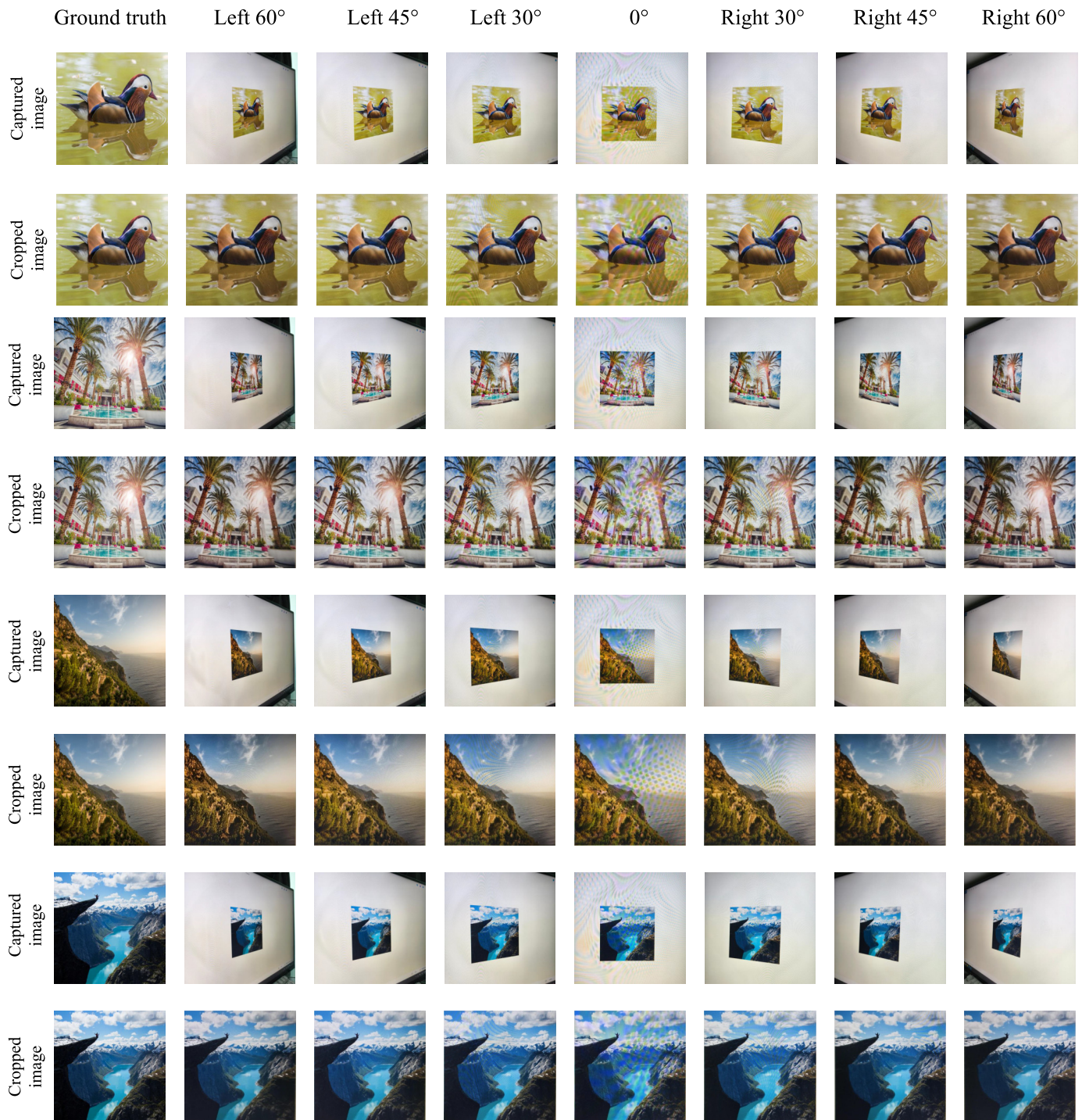


Fig. 11. (Continued.) Shooting environment and pictures at different angles.

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